

JESS Thermodynamic Database v8.9

Scandium

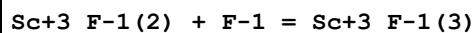
24-Jan-23

Reaction No. 2454							
F-1 + Sc+3 = Sc+3_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:6.19 (3SF)	4	MRX	140
2	25	0	Inf. Dilution	lgK:7.08 (3SF)	4	MRX	140
3	25	0	Inf. Dilution	lgK:7.08 (3SF)	5	CRV	22551
4	25	0.1	Unknown	lgK:6.22 (3SF)	4	CRV	552
5	25	0.5	NaClO4	lgK:6.18 (3SF)	4	MRX	140
6	25	0.5	NaClO4	lgK:6.19 (3SF)	4	MRX	140
7	25	0.5	NaClO4	lgK:6.22 (3SF)	5	MIS	309
8	25	0.5	NaClO4	lgK:6.2 (0.05SD)	5	CMN	437
9	25	0.5	NaClO4	lgK:6.17 (3SF)	5	MDS	5398
10	25	0.5	Unknown	lgK:6.18 (3SF)	5	EMU	5398
11	25	0.5	Unknown	lgK:6.18 (3SF)	5	CRV	6761
12	35	0.5	NaClO4	lgK:6.21 (3SF)	4	MRX	140
13	10:40	0.5	Unknown	dH:0.0 (1SF)	6	CRV	552
14	25	0.5	NaClO4	dH:0.41 (1.SD)	5	EVK	437
15	25	0.5	NaClO4	dH:0.406 (3SF)	3	MTD	437
16	25	0.5	NaClO4	dH:0.6 (1SF)	4	MRX	931

Reaction No. 2655							
Sc+3_F-1 + F-1 = Sc+3_F-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:5.32 (3SF)	4	MRX	140
2	25	0	Inf. Dilution	lgK:5.80 (3SF)	4	MRX	140
3	25	0	Inf. Dilution	lgK:5.81 (3SF)	4	MRX	140
4	25	0	Inf. Dilution	lgK:5.81 (3SF)	5	CRV	22551
5	25	0.5	NaClO4	lgK:5.27 (3SF)	4	MRX	140
6	25	0.5	NaClO4	lgK:5.11 (3SF)	4	MIS	309
7	25	0.5	NaClO4	lgK:5.2 (0.05SD)	5	CMN	437
8	25	0.5	NaClO4	lgK:5.28 (3SF)	4	MDS	437

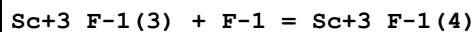
9	25	0.5	NaClO ₄	lgK:5.28 (3SF)	4	MRX	437
10	35	0.5	NaClO ₄	lgK:5.26 (3SF)	4	MRX	140
11	25	0.5	NaClO ₄	dH:-1.23 (3SF)	4	MTD	140
12	25	0.5	NaClO ₄	dH:-1. (1.SD)	5	EVK	437
13	25	0.5	NaClO ₄	dH:-1.195 (4SF)	3	MTD	437

Reaction No. 2660



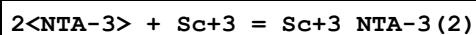
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO ₄	lgK:4.11 (3SF)	4	MRX	140
2	25	0	Inf. Dilution	lgK:4.45 (3SF)	4	MRX	140
3	25	0	Inf. Dilution	lgK:4.48 (3SF)	4	MRX	140
4	25	0	Inf. Dilution	lgK:4.48 (3SF)	5	CRV	22551
5	25	0.5	NaClO ₄	lgK:4.07 (3SF)	4	MRX	140
6	25	0.5	NaClO ₄	lgK:3.90 (3SF)	3	MIS	309
7	25	0.5	NaClO ₄	lgK:4.0 (0.05SD)	3	CMN	437
8	25	0.5	NaClO ₄	lgK:4.02 (3SF)	4	MDS	437
9	25	0.5	NaClO ₄	lgK:4.08 (3SF)	4	MRX	437
10	35	0.5	NaClO ₄	lgK:4.06 (3SF)	4	MRX	140
11	25	0.5	NaClO ₄	dH:-1. (1.SD)	5	EVK	437
12	25	0.5	NaClO ₄	dH:-1.267 (4SF)	3	MTD	437

Reaction No. 2661



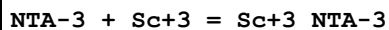
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO ₄	lgK:2.33 (3SF)	4	MRX	140
2	25	0	Inf. Dilution	lgK:3.48 (3SF)	0	MRX	140
3	25	0	Inf. Dilution	lgK:2.85 (3SF)	4	MRX	140
4	25	0	Inf. Dilution	lgK:2.85 (3SF)	5	CRV	22551
5	25	0.5	NaClO ₄	lgK:2.51 (3SF)	0	MRX	140
6	25	0.5	NaClO ₄	lgK:2.9 (0.05SD)	5	CMN	437
7	25	0.5	NaClO ₄	lgK:3.03 (3SF)	3	MDS	437
8	25	0.5	NaClO ₄	lgK:2.85 (3SF)	4	MRX	437
9	35	0.5	NaClO ₄	lgK:2.00 (3SF)	0	MRX	140

Reaction No. 3089



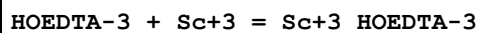
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KClO4	lgK:24.1(3SF)	4	MGL	1118
2	25	0	Inf. Dilution	lgK:24.4(3SF)	1	ESC	

Reaction No. 3090



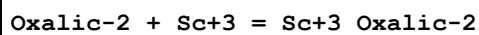
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:13.91(4SF)	0	MPL	1118
Note (1): Low wgt for internal consistency							
2	25	0.1	NaClO4	lgK:12.68(0.02SD)	6	MGL	188

Reaction No. 3153



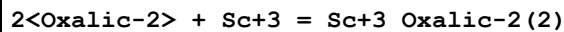
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:17.3(3SF)	6	CRV	552

Reaction No. 3707



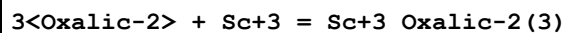
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:8.3(0.1SD)	0	MSP	4435
2	20	0.01	HNO3	lgK:9.4(0.05SD)	0	MSP	906
3	20	0.1	NaClO4	lgK:8.12(3SF)	5	MGL	6788
4	20	0.1	Unknown	lgK:8.30(3SF)	6	ENS	1539
5	20	1	Na+1 salt	lgK:6.86(3SF)	3	ENS	773
6	20	1	NaClO4	lgK:7.14(3SF)	3	MDS	906
7	25	0.05	Unknown	lgK:8.74(0.04SD)	4	MGL	906
8	25	0.05	Unknown	lgK:9.31(0.09SD)	2	MHG	906

Reaction No. 3708



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Na+1 salt	lgK:11.3(3SF)	0	ENS	773
2	20	1	NaClO4	lgK:12.69(4SF)	5	MDS	906
3	25	0	Inf. Dilution	lgK:15.1(3SF)	2	EAE	

Reaction No. 3709



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KClO4	lgK:16.28 (0.05SD)	4	MDS	931
2	20	0.1	Unknown	lgK:16.30 (4SF)	6	ENS	1539
3	20	1	Na+1 salt	lgK:14.32 (4SF)	0	ENS	773
4	20	1	NaClO4	lgK:15.68 (4SF)	5	MDS	906
5	25	0	Inf. Dilution	lgK:17.5 (3SF)	2	EAE	

Reaction No. 3710							
4<Oxalic-2> + Sc+3 = Sc+3_Oxalic-2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Na+1 salt	lgK:16.70 (4SF)	6	ENS	773
2	20	1	NaClO4	lgK:17.11 (4SF)	6	MDS	999
3	25	0	Inf. Dilution	lgK:16.9 (3SF)	2	EAE	

Reaction No. 3819							
Malonic-2 + Sc+3 = Sc+3_Malonic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:4.2 (0.06SD)	0	MSP	4435
Note (1): Low wgt for internal consistency							
2	25	0.1	KNO3	lgK:6.78 (3SF)	4	MGL	6793
3	25	1	NaClO4	lgK:5.87 (3SF)	5	MGL	999
4	25	1	Unknown	lgK:5.87 (3SF)	2	CRV	7271
5	25	1	Unknown	dH:3.4 (2SF)	3	CRV	7271

Reaction No. 3820							
2<Malonic-2> + Sc+3 = Sc+3_Malonic-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:11.90 (4SF)	5	MGL	6793
2	25	1	Na+1 salt	lgK:10.12 (4SF)	2	CRV	7271
3	25	1	NaClO4	lgK:10.12 (4SF)	6	MGL	999
4	25	1	Unknown	dH:6.3 (2SF)	2	CRV	7271

Reaction No. 3821							
3<Malonic-2> + Sc+3 = Sc+3_Malonic-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Na+1 salt	lgK:13.07 (4SF)	2	CRV	7271
2	25	1	NaClO4	lgK:13.07 (4SF)	6	MGL	999

3	25	1	Na+1 salt	dH:7.8 (2SF)	2	CRV	7271
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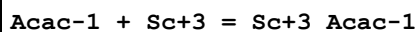
Reaction No. 3853							
Lactic-1 + Sc+3 = Sc+3_Lactic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:3.77 (3SF)	0	MGL	6788
2	25	0	Inf. Dilution	lgK:5.18 (3SF)	5	MPT	906
3	25	0.05	NaClO4	lgK:4.7 (0.2SD)	4	MPT	906
4	25	0	Ethanol:Water 19.8:80.2Wgt Inf. Dilution	lgK:5.89 (3SF)	5	MPT	906
5	25	0.05	Ethanol:Water 19.8:80.2Wgt NaClO4	lgK:5.3 (0.1SD)	5	MPT	906
6	25	0	Ethanol:Water 40.3:59.7Wgt Inf. Dilution	lgK:6.52 (3SF)	5	MPT	906
7	25	0.05	Ethanol:Water 40.3:59.7Wgt NaClO4	lgK:5.7 (0.1SD)	5	MPT	906

Reaction No. 4187							
EDTA-4 + Sc+3 = Sc+3_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.05	NaCl	lgK:21.84 (4SF)	0	MSP	906
2	20	0.1	KClO4	lgK:23.0 (3SF)	3	MDS	903
3	20	0.1	KNO3	lgK:23.1 (0.2SD)	3	MPL	136
4	20	0.1	NaClO4	lgK:24.13 (4SF)	0	MRX	142
5	20	0.1	NaClO4	lgK:25.05 (4SF)	0	MRX	931
Note (5): Low wgt for internal consistency							
6	20	0.1	NaClO4	lgK:21.00 (4SF)	0	MGL	6788
7	25	0	Inf. Dilution	lgK:25.5 (3SF)	2	EAE	

Reaction No. 4188							
Sc+3_EDTA-4 + H+1 = Sc+3_H+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:10.88 (4SF)	0	MPL	140
2	20	0.1	NaClO4	lgK:2. (1SF)	3	MRX	142
3	20	0.1	NaClO4	lgK:2. (1SF)	7	EIU	903
4	20	0.1	Unknown	lgK:3.29 (3SF)	0	ENS	773

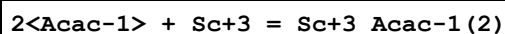
5	25	0	Inf. Dilution	lgK:2.21 (3SF)	2	EAE	
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Reaction No. 4531



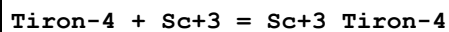
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:8.4 (0.1SD)	4	MSP	4435
2	20	0.1	Unknown	lgK:8.00 (3SF)	6	ENS	1539
3	25	0.1	NaClO4	lgK:8.3 (2SF)	4	MDS	999
4	30	0	Inf. Dilution	lgK:8.0 (2SF)	4	MGL	999

Reaction No. 4532



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:15.2 (3SF)	4	ENS	1539
2	25	0	Inf. Dilution	lgK:15.2 (3SF)	1	ESC	
3	30	0	Unknown	lgK:15.2 (3SF)	4	ENS	773

Reaction No. 4636

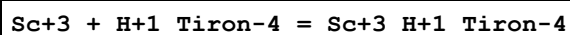


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO4	lgK:18.96 (4SF)	5	MGL	999
2	25	0.1	KNO3	lgK:18.07 (4SF)	0	MGL	999

Note (2): Low wgt for internal consistency

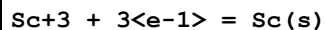
3	25	0.1	NaClO4	lgK:18.96 (4SF)	5	MGL	999
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Reaction No. 4637



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.94 (3SF)	3	MGL	999

Reaction No. 4875



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:140.2 (4SF)	3	CRV	6488
2	25	0	Inf. Dilution	dG:586.6 (4SF) kJ	5	CRV	7421
3	25	0	Inf. Dilution	dG:140.200 (6SF)	0	CRV	9029
4	25	0	Inf. Dilution	dG:140.20 (5SF)	3	CRV	10174

5	25	0	Inf. Dilution	dG:586.6 (4SF) kJ	4	EFE	32152
6	50	0	Inf. Dilution	dG:138.64 (5SF)	3	CRV	10174
7	75	0	Inf. Dilution	dG:137.03 (5SF)	2	CRV	10174
8	100	0	Inf. Dilution	dG:135.37 (5SF)	2	CRV	10174
9	125	0	Inf. Dilution	dG:133.65 (5SF)	2	CRV	10174
10	150	0	Inf. Dilution	dG:131.88 (5SF)	3	CRV	10174
11	175	0	Inf. Dilution	dG:130.04 (5SF)	3	CRV	10174
12	200	0	Inf. Dilution	dG:128.13 (5SF)	1	CRV	10174
13	225	0	Inf. Dilution	dG:126.14 (5SF)	0	CRV	10174
14	250	0	Inf. Dilution	dG:124.05 (5SF)	0	CRV	10174
15	300	0	Inf. Dilution	dG:119.54 (5SF)	0	CRV	10174
16	25	0	Inf. Dilution	E:-2.1 (0.1SD)	0	MDG	775
17	25	0	Inf. Dilution	E:-2.077 (4SF)	0	CRV	6330
18	25	0	Inf. Dilution	E:-2.09 (3SF)	0	CRV	7761
19	25	0	Inf. Dilution	E:-2.077 (4SF)	0	CRV	22519
20	25	0	Inf. Dilution	E:-2.03 (3SF)	1	CRV	22551
21	25	0	Inf. Dilution	dH:146.8 (4SF)	0	CRV	6488
22	25	0	Inf. Dilution	dH:640.3 (4SF) kJ	0	CRV	7761
23	25	0	Inf. Dilution	dH:146.800 (6SF)	0	CRV	9029
24	25	0	Inf. Dilution	dH:614.2 (4SF) kJ	4	EFE	32152

Reaction No. 4918

IDA-2 + Sc+3 = Sc+3_IDA-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaCl	lgK:10.10 (4SF)	3	MGL	2261
2	20	0.1	NaClO4	lgK:9.80 (3SF)	3	MGL	6788
3	25	0.1	KCl	lgK:9.35 (3SF)	0	MGL	2261
4	25	1	Na+1 salt	lgK:15.9 (3SF)	0	CRV	552
5	25	1	NaClO4	lgK:9.85 (3SF)	4	MGL	999
6	20	0.1	Water:Ethanol 40:60Wgt NaCl	lgK:11.47 (4SF)	5	MGL	2261

Reaction No. 4919

2<IDA-2> + Sc+3 = Sc+3_IDA-2 (2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:16.06 (4SF)	6	MGL	999

Reaction No. 4920							
Sc+3 + H+1 + IDA-2 = Sc+3_H+1_IDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.6(3SF)	1	ELC	
2	25	1	NaClO4	lgK:11.59(4SF)	4	MGL	999
3	25	1	Unknown	lgK:10.6(3SF)	2	ENS	773

Reaction No. 4921							
Sc+3 + 2<H+1> + IDA-2 = Sc+3_H+1(2)_IDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:13.5(3SF)	4	MGL	999

Reaction No. 5178							
Cat-2 + Sc+3 = Sc+3_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:17.0(0.03SD)	4	MGL	999

Reaction No. 6438							
Sc+3 + H+1_Oxalic-2 = Sc+3_H+1_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.05	Unknown	lgK:7.36(0.04SD)	4	MGL	906

Reaction No. 9181							
Sc+3_NTA-3 + OH-1 = Sc+3_OH-1_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	KNO3	lgK:7.44(0.01SD)	4	MGL	1118
2	25	0	Inf. Dilution	lgK:7.6(2SF)	1	ESC	

Reaction No. 10493							
Succinic-2 + Sc+3 = Sc+3_Succinic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:4.5(0.04SD)	0	MSP	4435
Note (1): Low wgt for internal consistency							
2	25	0.1	KNO3	lgK:5.55(3SF)	4	MGL	6793

Reaction No. 10496							
Sc+2 + H+1_Succinic-2 = Sc+2_H+1_Succinic-2							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	KCl	lgK:0.48 (2SF)	6	MHY	999

Reaction No. 10543

Diglycol-2 + Sc+3 = Sc+3_Diglycol-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:8.3 (0.2SD)	3	MCL	999

Reaction No. 10544

Sc+3_Diglycol-2 + Diglycol-2 = Sc+3_Diglycol-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:4.49 (0.02SD)	6	MGL	999

Reaction No. 10657

2<Tartaric-2> + Sc+3 = Sc+3_Tartaric-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KClO4	lgK:12.5 (0.1SD)	4	MDS	999

Reaction No. 11436

Sc+3_Acac-1 + Acac-1 = Sc+3_Acac-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.4 (2SF)	1	ESC	
2	30	0	Inf. Dilution	lgK:7.2 (2SF)	4	MGL	999

Reaction No. 11488

Glutaric-2 + Sc+3 = Sc+3_Glutaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:4.0 (0.2SD)	0	MSP	4435

Note (1): Low wgt for internal consistency

2	25	0.05	Unknown	lgK:4.82 (0.02SD)	3	MGL	906
3	25	0.1	KNO3	lgK:5.21 (3SF)	4	MGL	6793
4	25	0.05	Ethanol:Water 25:75Vol Unknown	lgK:5.6 (0.04SD)	4	MGL	906
5	25	0.05	Ethanol:Water 40:60Vol Unknown	lgK:6.24 (0.02SD)	4	MGL	906
6	25	0.05	Ethanol:Water 50:50Vol Unknown	lgK:6.96 (0.01SD)	4	MGL	906

Reaction No. 11782							
Phenol-1 + Sc+3 = Sc+3_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:7.2 (0.07SD)	4	MGL	999

Reaction No. 11808							
Sc+3 + H+1(2)_Cat-2 = Sc+3_Cat-2 + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5. (1SF)	1	EES	
2	25	0.1	Unknown	lgK:-4.39 (3SF)	0	MGL	999

Reaction No. 12564							
Sc+3 + OH-1 + NTA-3 = Sc+3_OH-1_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.2	Unknown	lgK:7.44 (0.01SD)	0	MGL	906
2	25	0	Inf. Dilution	lgK:22.0 (3SF)	1	ELC	

Reaction No. 12588							
Sc+3_Tiron-4 + H2O = Sc+3_OH-1_Tiron-4 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-6.10 (3SF)	6	MGL	999

Reaction No. 12774							
HIMDA-2 + Sc+3 = Sc+3_HIMDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaNO3	lgK:8.65 (3SF)	6	MSC	999

Reaction No. 13321							
26PyridineDiCOO-2 + Sc+3 = Sc+3_26PyridineDiCOO-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:11.2 (0.2SD)	3	MCL	999

Reaction No. 13322							
Sc+3_26PyridineDiCOO-2 + 26PyridineDiCOO-2 = Sc+3_26PyridineDiCOO-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:7.66 (0.01SD)	6	MHG	999

Reaction No. 13529							
Mandelic-1 + Sc+3 = Sc+3_Mandelic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:5.4 (0.06SD)	0	MSP	4435
Note (1): Low wgt for internal consistency							
2	25	0.1	KNO3	lgK:2.91 (3SF)	4	MGL	7703

Reaction No. 14021							
Sc+3_OH-1_EDTA-4 + H+1 = Sc+3_EDTA-4 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:10.54 (4SF)	5	MPL	140
2	20	0.1	NaClO4	lgK:10.66 (4SF)	4	MRX	142
3	20	0.1	Unknown	lgK:10.70 (4SF)	6	ENS	1539
4	25	0	Inf. Dilution	lgK:11.0 (3SF)	2	EAE	
5	40	0.1	Unknown	lgK:10.6 (0.2SD)	3	MMR	906

Reaction No. 14048							
Sc+3 + EDTA-4 + OH-1 = Sc+3_OH-1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KClO4	lgK:27.4 (0.03SD)	0	MDS	903
2	25	0	Inf. Dilution	lgK:28.5 (3SF)	1	ELC	

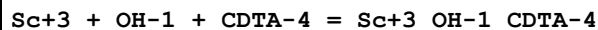
Reaction No. 14049							
Sc+3_EDTA-4 + OH-1 = Sc+3_OH-1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:3.3 (2SF)	3	MRX	931
2	25	0	Inf. Dilution	lgK:4.1 (2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:3.09 (3SF)	0	EAE	

Reaction No. 14252							
Sc+3 + H+1_PAR-2 = Sc+3_H+1_PAR-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KCl	lgK:6.25 (3SF)	6	MSP	999

Reaction No. 14662							
CDTA-4 + Sc+3 = Sc+3_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

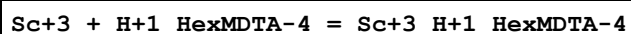
1	20	0.1	KClO ₄	lgK:25.4 (3SF)	3	MDS	999
2	20	0.1	Unknown	lgK:26.10 (4SF)	3	ENS	1539
3	25	0	Inf. Dilution	lgK:25.8 (3SF)	1	ESC	

Reaction No. 14663



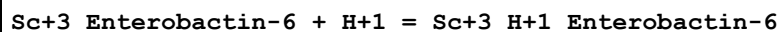
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KClO ₄	lgK:28.0 (3SF)	4	MDS	999
2	25	0	Inf. Dilution	lgK:28.4 (3SF)	1	ESC	

Reaction No. 14877



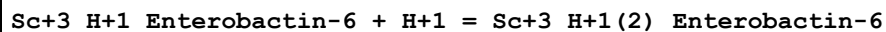
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	KNO ₃	lgK:9.68 (3SF)	6	MSP	999

Reaction No. 16153



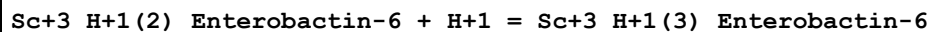
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:4.67 (3SF)	6	MGL	1302

Reaction No. 16154



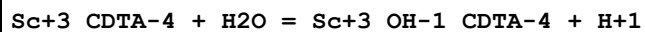
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:3.84 (3SF)	6	MGL	1302

Reaction No. 16155



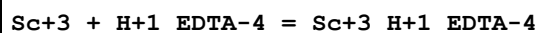
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:3.31 (3SF)	6	MGL	1302

Reaction No. 18014



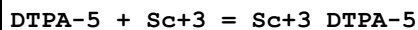
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-11.40 (4SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:-11.2 (3SF)	1	ESC	

Reaction No. 18075



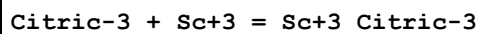
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:14.90 (4SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:16.8 (3SF)	2	EAE	

Reaction No. 18107



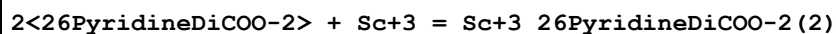
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22			lgK:26. (0.4SD)	4	MSP	906
2	18:22			lgK:27. (0.4SD)	4	MSP	906
3	20	0.1	Unknown	lgK:24.50 (4SF)	1	ENS	1539
4	25	0	Inf. Dilution	lgK:29.6 (3SF)	2	EAE	
5	25	0.1	NaCl	lgK:26.6 (0.2SD)	3	MPH	31393
6	27	0.1	Unknown	dH:-7.6 (2SF)	3	CRV	2556

Reaction No. 18175



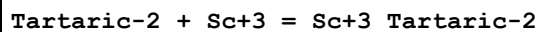
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:7.00 (3SF)	0	MSP	4435
2	20	0.1	Unknown	lgK:7.00 (3SF)	3	ENS	1539
3	25	0	Inf. Dilution	lgK:8.5 (2SF)	1	EDH	

Reaction No. 18252



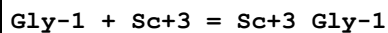
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:18.90 (4SF)	6	ENS	1539

Reaction No. 18318



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:6.2 (0.06SD)	4	MSP	4435
2	20	0.1	Unknown	lgK:6.20 (3SF)	6	ENS	1539

Reaction No. 18536



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaCl	lgK:6.95 (3SF)	5	MGL	1555
2	20	0.1	NaClO4	lgK:7.13 (3SF)	4	MGL	6788

3	25	0	Inf. Dilution	lgK:8.08 (3SF)	2	EAE	
4	30	0.2	NaClO ₄	lgK:7.75 (3SF)	0	MGL	11516
Note (4): Makhijani S et al., J. Indian Chem. Soc., 1977, 54, 670							

Reaction No. 20730							
Sc+3_CDTA-4 + OH-1 = Sc+3_OH-1_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:2.6 (2SF)	4	ENS	773
2	25	0	Inf. Dilution	lgK:2.6 (2SF)	1	ESC	

Reaction No. 21555							
2ThenTriFAcone-1 + Sc+3 = Sc+3_2ThenTriFAcone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO ₄	lgK:6.34 (3SF)	0	MDS	999
2	25	0.1	NaClO ₄	lgK:7.1 (2SF)	3	MDS	999

Reaction No. 23025							
Sc+3_Tiron-4 + H+1 = Sc+3_H+1_Tiron-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.6 (2SF)	1	ELC	
2	25	0.1	KNO ₃	lgK:-1.92 (3SF)	0	MGL	999

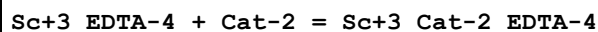
Reaction No. 23543							
2<2ThenTriFAcone-1> + Sc+3 = Sc+3_2ThenTriFAcone-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO ₄	lgK:11.76 (4SF)	6	MDS	999
2	25	0.1	NaClO ₄	lgK:11.76 (4SF)	6	MDS	999

Reaction No. 23544							
3<2ThenTriFAcone-1> + Sc+3 = Sc+3_2ThenTriFAcone-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO ₄	lgK:19.26 (4SF)	6	MDS	999
2	25	0.1	NaClO ₄	lgK:19.26 (4SF)	6	MDS	999

Reaction No. 23717							
Sc+3_EDTA-4 + Tiron-4 = Sc+3_EDTA-4_Tiron-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

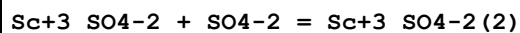
1	25	0.1	KNO3	lgK:8.5(0.2SD)	3	MGL	906
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Reaction No. 23718



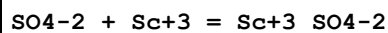
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.1(0.2SD)	3	MGL	906

Reaction No. 24684



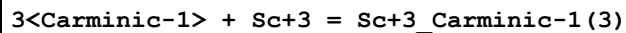
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.66(3SF)	5	MCL	5398
2	25	0	Inf. Dilution	lgK:1.66(0.04SD)	4	MCL	8402
3	25	0.5	NaClO4	lgK:1.38(3SF)	5	MIX	11516
Note (3): Tateda A, Bull. Chem. Soc. Jpn., 1965, 38, 165							
4	25	0.5	NaClO4	lgK:1.37(3SF)	5	MIX	15217
5	25	0	Inf. Dilution	dH:3.96(3SF)	5	MCL	5398
6	25	0	Inf. Dilution	dH:3.96(0.10SD)	4	MCL	8402

Reaction No. 24685



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:2.57(3SF)	5	MCE	931
2	23	0.5	NaClO4	lgK:1.66(3SF)	0	MCE	931
3	23	2	NaClO4	lgK:0.62(2SF)	0	MCE	931
4	25	0	Inf. Dilution	lgK:4.20(3SF)	4	CRV	552
5	25	0	Inf. Dilution	lgK:4.04(3SF)	4	MCL	5398
6	25	0	Inf. Dilution	lgK:4.04(0.03SD)	4	MCL	8402
7	25	0	Inf. Dilution	lgK:4.04(3SF)	5	EOD	22958
8	25	0.5	NaClO4	lgK:2.59(3SF)	3	MCE	931
9	25	0	Inf. Dilution	dH:7.5(2SF)	3	CRV	552
10	25	0	Inf. Dilution	dH:6.31(3SF)	4	MCL	5398
11	25	0	Inf. Dilution	dH:6.31(0.03SD)	4	MCL	8402

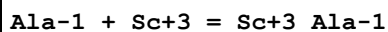
Reaction No. 25482



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	Mixed salts	lgK:16.32(4SF)	0	MPL	2261

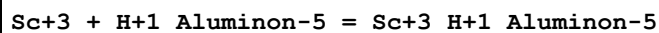
Note (1): Thermodynamically isolated

Reaction No. 29182



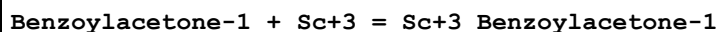
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:7.7(0.2SD)	3	MGL	2392

Reaction No. 37255



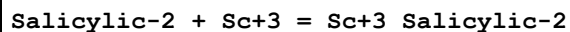
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:4.5(0.2SD)	3	MSP	931

Reaction No. 37470



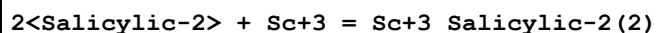
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:8.8(0.2SD)	4	MSP	4435

Reaction No. 38179



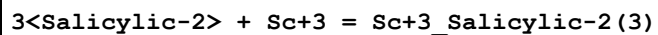
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:14.20(4SF)	3	MGL	6788
2	25	0	Inf. Dilution	lgK:15.4(3SF)	2	EAE	
3	23	0.25	Acetone:Water 50:50Vol NaClO4	lgK:3.23(3SF)	4	MCE	1614

Reaction No. 38180



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.25	Acetone:Water 50:50Vol NaClO4	lgK:5.920(0.002SD)	4	MCE	1614

Reaction No. 38181



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.25	Acetone:Water 50:50Vol NaClO4	lgK:9.(0.3SD)	4	MCE	1614

Reaction No. 40174							
Sc+3 + H+1_Salicylic-2 = Sc+3_H+1_Salicylic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaCl	lgK:2.48 (3SF)	5	MGL	2261
2	25	0	Inf. Dilution	lgK:3.1 (3SF)	2	EAE	

Reaction No. 40175							
Sc+3 + 2<H+1_Salicylic-2> = Sc+3_H+1(2)_Salicylic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaCl	lgK:4.70 (3SF)	5	MGL	2261
2	25	0	Inf. Dilution	lgK:5.73 (3SF)	2	EAE	

Reaction No. 40176							
Sc+3 + 3<H+1_Salicylic-2> = Sc+3_H+1(3)_Salicylic-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaCl	lgK:6.72 (3SF)	5	MGL	2261
2	25	0	Inf. Dilution	lgK:7.96 (3SF)	2	EAE	

Reaction No. 40760							
Adipic-2 + Sc+3 = Sc+3_Adipic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HNO3	lgK:4.4 (0.2SD)	0	MSP	4435
Note (1): Low wgt for internal consistency							
2	25	0.1	KNO3	lgK:5.25 (3SF)	5	MGL	6793

Reaction No. 41728							
SulfSal-3 + Sc+3 = Sc+3_SulfSal-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.01	HCl	lgK:4.0 (0.08SD)	0	MSP	4435
2	20	0.1	NaClO4	lgK:11.23 (4SF)	5	MGL	6788
3	25	0	Inf. Dilution	lgK:11.4 (3SF)	1	ESC	

Reaction No. 43581							
234TriOHPentDioic-2 + Sc+3 = Sc+3_234TriOHPentDioic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:5.4 (0.1SD)	4	MPT	906

Reaction No. 44651							
3<Oxine-1> + Sc+3 = Sc+3_Oxine-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Dioxan:Water 60:40Wgt NaClO4	lgK:30.54(4SF)	4	MGL	3082

Reaction No. 44879							
Sc+3_EDTA-4 + 23DiOH6SulfNaphthoic-3 = Sc+3_23DiOH6SulfNaphthoic-3_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.3(0.2SD)	4	MGL	906

Reaction No. 45424							
Sc+3 + 3<Cl-1> = Sc+3_Cl-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.24(4SF)	1	ELC	

Reaction No. 45687							
Sc+3 + H+1(2)_XylenolOrange-6 = Sc+3_H+1(2)_XylenolOrange-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.5(2SF)	4	MSP	906

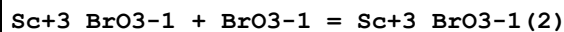
Reaction No. 45712							
Sc+3 + H+1(3)_MeThymolBlue-6 = Sc+3_H+1(3)_MeThymolBlue-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:12.5(0.03SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:12.7(3SF)	1	ESC	

Reaction No. 45713							
Sc+3 + H+1(2)_MeThymolBlue-6 = Sc+3_H+1(2)_MeThymolBlue-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:16.55(4SF)	4	MSP	906
2	25	0	Inf. Dilution	lgK:16.8(3SF)	1	ESC	

Reaction No. 52943							
BrO3-1 + Sc+3 = Sc+3_BrO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	1	HClO4	lgK:0.79(2SF)	5	MDS	5398

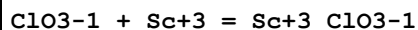
2	20	1	HClO4	lgK:0.76 (2SF)	5	MDS	5398
3	25	1	HClO4	lgK:0.65 (2SF)	5	MDS	5398
4	30	1	HClO4	lgK:0.56 (2SF)	5	MDS	5398
5	10:40	1	Unknown	dH:-8. (1SF)	6	CRV	552

Reaction No. 52944



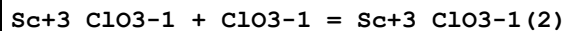
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	1	HClO4	lgK:-0.23 (2SF)	5	MDS	5398
2	20	1	HClO4	lgK:-0.05 (1SF)	5	MDS	5398
3	25	1	HClO4	lgK:0.09 (1SF)	5	MDS	5398
4	30	1	HClO4	lgK:0.22 (2SF)	5	MDS	5398

Reaction No. 52957



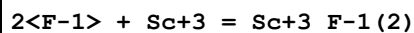
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15:30			lgK:-0.02 (1SF)	0	MDS	5398
2	15:30			dH:0.0 (1SF)	0	MTD	5398

Reaction No. 52958



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15			lgK:-0.80 (2SF)	0	MDS	5398
2	20			lgK:-0.89 (2SF)	0	MDS	5398
3	25			lgK:-1.10 (3SF)	0	MDS	5398
4	30			lgK:-1.30 (3SF)	0	MDS	5398

Reaction No. 53148



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.9 (3SF)	5	CRV	2556
2	25	0.1	Unknown	lgK:11.5 (3SF)	5	CRV	2556
3	25	0.5	NaClO4	lgK:11.33 (4SF)	5	MIS	309
4	25	0.5	NaClO4	lgK:11.44 (4SF)	5	MDS	5398
5	0:40	0.5	Unknown	dH:-1. (1SF)	5	CRV	2556

Reaction No. 53149

3<F-1> + Sc+3 = Sc+3_F-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.4(3SF)	5	CRV	2556
2	25	0.1	Unknown	lgK:15.5(3SF)	4	CRV	2556
3	25	0.5	NaClO4	lgK:15.23(4SF)	5	MIS	309
4	25	0.5	NaClO4	lgK:15.46(4SF)	5	MDS	5398
5	0:40	0.5	Unknown	dH:-2.(1SF)	5	CRV	2556

Reaction No. 53150							
4<F-1> + Sc+3 = Sc+3_F-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.3(3SF)	5	CRV	2556
2	25	0.5	NaClO4	lgK:18.49(4SF)	5	MDS	5398

Reaction No. 53365							
Sc+3 + H+1(2)_PO4-3 = Sc+3_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.72(3SF)	5	MCE	5398
2	25	0.6	NaClO4	lgK:3.76(3SF)	5	MCE	5398

Reaction No. 53366							
2<Sc+3> + H+1(3)_PO4-3 = Sc+3(2)_H+1_PO4-3 + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	NH4NO3	lgK:3.53(3SF)	4	MCE	5398
2	25	0	Inf. Dilution	lgK:3.53(3SF)	2	EAE	

Reaction No. 53621							
NO3-1 + Sc+3 = Sc+3_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:0.34(2SF)	3	MCE	5398
2	23	2	NaClO4	lgK:0.55(2SF)	0	MCE	931
3	25	0	Inf. Dilution	lgK:0.8(1SF)	4	EAC	
4	25	4	HClO4	lgK:0.28(2SF)	0	MCE	931

Reaction No. 53829							
2<NO3-1> + Sc+3 = Sc+3_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	23	2	NaClO4	lgK:0.08 (1SF)	1	MCE	931
2	25	4	HClO4	lgK:-0.28 (2SF)	2	MCE	931

Reaction No. 54026							
2<SO4-2> + Sc+3 = Sc+3_SO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	NaClO4	lgK:3.04 (3SF)	0	MCE	931
2	23	2	NaClO4	lgK:1.50 (3SF)	0	MCE	931
3	25	0	Inf. Dilution	lgK:5.7 (2SF)	3	CRV	2556
4	25	0.5	NaClO4	lgK:3.96 (3SF)	4	MCE	931
5	25	0	Inf. Dilution	dH:13. (2SF)	3	CRV	2556

Reaction No. 54027							
3<SO4-2> + Sc+3 = Sc+3_SO4-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	NaClO4	lgK:4.0 (2SF)	4	MCE	931
2	25	0	Inf. Dilution	lgK:5.77 (3SF)	2	EAE	

Reaction No. 54028							
Sc+3 + H+1_SO4-2 = Sc+3_SO4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.67 (3SF)	2	EAE	
2	20	1	HClO4	lgR:1.49 (3SF)	5	MCE	931

Reaction No. 54103							
Sc+3 + SeO4-2 = Sc+3_SeO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:1.78 (0.05SD)	5	MIX	931
2	35	0.5	NaClO4	lgK:1.75 (3SF)	5	MCE	931
3	45	0.5	NaClO4	lgK:1.68 (3SF)	5	MCE	931
4	25:45	0.5	NaClO4	dH:-2.3 (0.05SD)	5	MIX	931

Reaction No. 54104							
Sc+3 + 2<SeO4-2> = Sc+3_SeO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:2.64 (0.05SD)	2	MIX	931
2	35	0.5	NaClO4	lgK:3.1 (0.05SD)	2	MIX	931

3	45	0.5	NaClO4	lgK:3.3 (0.05SD)	0	MIX	931
4	25:45	0.5	NaClO4	dH:39.2 (3SF)	2	MTD	931

Reaction No. 54158							
Sc+3_Cl-1 + Cl-1 = Sc+3_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:1.11 (3SF)	4	MAG	140
2	20	0.691	HClO4	lgK:-0.15 (2SF)	0	MCE	931
Note (2): Low wgt for internal consistency							
3	25	0.5	NaClO4	lgK:1.04 (3SF)	4	MAG	140
4	35	0.5	NaClO4	lgK:0.91 (2SF)	4	MAG	140

Reaction No. 54159							
Cl-1 + Sc+3 = Sc+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:1.14 (3SF)	0	MAG	140
2	20	0.691	HClO4	lgK:0.04 (0.04SD)	0	MIX	5207
3	20	1	HClO4	lgK:-0.80 (2SF)	0	MCE	931
4	23	2	NaClO4	lgK:0.14 (2SF)	0	MCE	931
5	23	6	HClO4	lgK:1.45 (3SF)	4	MDS	931
6	25	0.5	NaClO4	lgK:1.07 (3SF)	3	MAG	140
7	25	4	HClO4	lgK:-0.12 (2SF)	0	MCE	931
8	35	0.5	NaClO4	lgK:1.00 (3SF)	3	MAG	140
9	25	0	Inf. Dilution	dG:-9.382 (4SF) kJ	1	EES	
Note (9): From DOCMN on GBL							

Reaction No. 54160							
Sc+3 + 2<Cl-1> = Sc+3_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.691	HClO4	lgK:-0.1 (0.06SD)	0	MIX	5207
2	23	2	NaClO4	lgK:0.54 (2SF)	0	MCE	931
3	23	6	HClO4	lgK:0.70 (2SF)	2	MDS	931
4	25	0	Inf. Dilution	lgK:3.6 (2SF)	1	ELC	
5	25	0	Inf. Dilution	lgK:3.4 (2SF)	0	EOR	
6	25	4	HClO4	lgK:-0.84 (2SF)	3	MCE	931
7	25	0	Inf. Dilution	dG:-14.71 (4SF) kJ	0	EOR	
Note (7): From DOCMN on GBL							

Reaction No. 54161							
$3\text{<Cl-1>} + \text{Sc+3} = \text{Sc+3_Cl-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:0.16(2SF)	4	MCE	931
2	23	6	HClO4	lgK:-0.22(2SF)	4	MDS	931
3	25	4	HClO4	lgK:-1.40(3SF)	4	MCE	931

Reaction No. 54162							
$4\text{<Cl-1>} + \text{Sc+3} = \text{Sc+3_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	6	HClO4	lgK:-0.37(2SF)	4	MDS	931
2	23	10	HClO4	lgK:-1.88(3SF)	4	MAX	931

Reaction No. 54279							
$\text{Sc+3_Br-1} + \text{Br-1} = \text{Sc+3_Br-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:0.54(2SF)	4	MAG	140
2	20	0.691	HClO4	lgK:-0.33(2SF)	0	MCE	931
3	25	0	Inf. Dilution	lgK:1.00(3SF)	4	MAG	140
4	25	0.5	NaClO4	lgK:0.52(2SF)	4	MAG	140
5	35	0.5	NaClO4	lgK:0.48(2SF)	4	MAG	140
6	25	0.5	NaClO4	dH:-0.6(1SF)	4	MTD	140

Reaction No. 55330							
$\text{Sc+3} + \text{H2O} = \text{Sc+3_OH-1} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.3(2SF)	0	CRV	6478
2	25	0	Unknown	lgK:-4.3(2SF)	2	EOD	31393
3	25	0.1	KNO3	lgK:-4.84(0.008SD)	0	MPS	4880
Note (3): EBP: $\text{Sc+3} + \text{OH-1} = \text{Sc+3_OH-1}$							
4	25	0	Inf. Dilution	dH:11.6(3SF)	1	CRV	6478
5	25	0	Inf. Dilution	dH:14.4(1.4SD)	1	CRV	6478

Reaction No. 55331							
$2\text{<Sc+3>} + 2\text{<H2O>} = \text{Sc+3(2)_OH-1(2)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0.1	KNO3	lgK:-6.096(0.004SD)	5	MPS	4880
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Reaction No. 55332							
$3\text{Sc} + 3\text{H}_2\text{O} = \text{Sc}_3\text{OH} + 3\text{H}_2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.9(3SF)	1	ELC	
2	25	0.1	KNO3	lgK:-17.57(0.006SD)	0	MPS	4880
Note (2): EBP: $3\text{Sc} + 3\text{OH} = \text{Sc}_3\text{OH}$							

Reaction No. 56383							
$1.5\text{O}_2(\text{g}) + 2\text{Sc}(\text{s}) = \text{Sc}_2\text{O}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:318.7(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:316.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:233.6(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:183.8(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:150.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-434.8(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-434.8(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-434.8(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dG:-1815.(4SF) kJ	5	CRV	30332
Note (9): Value from LW3JTH option 2							
10	25	0	Inf. Dilution	dG:-434.85(5SF)	3	CRV	31291
11	25	0	Inf. Dilution	dH:-456.2(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dH:-456.2(0.1SD)	5	ENS	802
13	25	0	Inf. Dilution	dH:-456.2(4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-1904.5(5SF) kJ	5	CRV	30332
15	127	0	Inf. Dilution	dH:-456.1(4SF)	2	MTD	6422
16	227	0	Inf. Dilution	dH:-455.8(4SF)	1	MTD	6422
17	327	0	Inf. Dilution	dH:-455.5(4SF)	0	MTD	6422

Reaction No. 56384							
$\text{Sc}(\text{s}) + 1.5\text{O}_2(\text{g}) + 1.5\text{H}_2(\text{g}) = \text{Sc}_3\text{OH} + 1.5\text{H}_2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-294.8(0.05SD)	0	ENS	802
Note (1): Low wgt for inter-reaction consistency							
2	25	0	Inf. Dilution	dH:-325.9(0.1SD)	0	ENS	802

Reaction No. 56385							
$\text{Sc(s)} + 1.5\text{F}_2\text{(g)} = \text{ScF}_3\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:269.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:268.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:197.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:155.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:127.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-371.8(0.05SD)	0	ENS	802
7	25	0	Inf. Dilution	dG:-386.0(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-389.4(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-389.4(0.1SD)	5	ENS	802
10	25	0	Inf. Dilution	dH:-385.2(4SF)	4	MTD	6422
11	127	0	Inf. Dilution	dH:-384.9(4SF)	4	MTD	6422
12	227	0	Inf. Dilution	dH:-384.5(4SF)	4	MTD	6422
13	327	0	Inf. Dilution	dH:-384.1(4SF)	4	MTD	6422

Reaction No. 56888							
$1.5\text{Cl}_2\text{(g)} + \text{Sc(s)} = \text{ScCl}_3\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:148.4(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:108.2(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:84.10(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:68.07(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-203.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-925.1(4SF) kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-221.1(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-220.7(4SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-220.2(4SF)	3	MTD	6422
11	327	0	Inf. Dilution	dH:-219.7(4SF)	0	MTD	6422

Reaction No. 57201							
$\text{Sc(s)} + 0.5\text{N}_2\text{(g)} = \text{ScN(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.71(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:49.37 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:35.71 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:27.52 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:22.06 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-67.82 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-75.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-75.0 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-74.9 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-74.9 (3SF)	4	MTD	6422

Reaction No. 57203							
2<Br-1> + Sc+3 = Sc+3_Br-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.691	HClO4	lgK:-0.4 (0.06SD)	3	MIX	5208
2	25	0	Inf. Dilution	lgK:3.2 (2SF)	1	ELC	

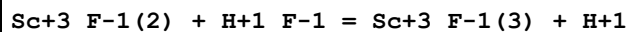
Reaction No. 57204							
Br-1 + Sc+3 = Sc+3_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:1.24 (3SF)	4	MAG	140
2	20	0.691	HClO4	lgK:-0.07 (0.05SD)	0	MIX	5208
3	25	0	Inf. Dilution	lgK:2.08 (3SF)	4	MAG	140
4	25	0.5	NaClO4	lgK:1.21 (3SF)	4	MAG	140
5	35	0.5	NaClO4	lgK:1.18 (3SF)	4	MAG	140
6	25	0.5	NaClO4	dH:-0.5 (1SF)	4	MTD	140

Reaction No. 58539							
Sc+3 + H+1_F-1 = Sc+3_F-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:3.33 (3SF)	4	MRX	140
2	25	0.5	NaClO4	lgK:3.28 (3SF)	4	MRX	140
3	35	0.5	NaClO4	lgK:3.21 (3SF)	4	MRX	140
4	25	0.5	NaClO4	dH:-2.3 (2SF)	4	MTD	140

Reaction No. 58540							
Sc+3_F-1 + H+1_F-1 = Sc+3_F-1(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

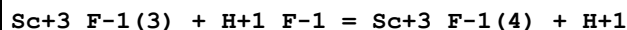
1	15	0.5	NaClO4	lgK:2.46 (3SF)	4	MRX	140
2	25	0.5	NaClO4	lgK:2.37 (3SF)	4	MRX	140
3	35	0.5	NaClO4	lgK:2.26 (3SF)	4	MRX	140
4	25	0.5	NaClO4	dH:-4.1 (2SF)	4	MTD	140

Reaction No. 58541



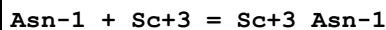
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:1.23 (3SF)	4	MRX	140
2	25	0.5	NaClO4	lgK:1.16 (3SF)	4	MRX	140
3	35	0.5	NaClO4	lgK:1.06 (3SF)	4	MRX	140
4	25	0.5	NaClO4	dH:-3.4 (2SF)	4	MTD	140

Reaction No. 58542



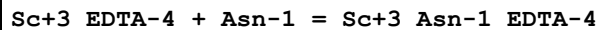
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	NaClO4	lgK:-0.15 (2SF)	4	MRX	140
2	25	0.5	NaClO4	lgK:-0.07 (1SF)	4	MRX	140
3	35	0.5	NaClO4	lgK:-0.30 (2SF)	4	MRX	140

Reaction No. 59569



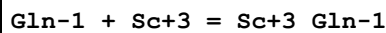
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.84 (3SF)	2	EAE	
2	30	0.2	NaClO4	lgK:7.083 (4SF)	4	MGL	905

Reaction No. 59593



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.13 (3SF)	2	EAE	
2	30	0.2	NaClO4	lgK:3.38 (3SF)	4	MGL	905

Reaction No. 59612



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.17 (3SF)	2	EAE	
2	30	0.2	NaClO4	lgK:7.405 (4SF)	4	MGL	905

Reaction No. 59631							
Sc+3_EDTA-4 + Gln-1 = Sc+3_Gln-1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.18 (3SF)	2	EAE	
2	30	0.2	NaClO4	lgK:3.43 (3SF)	4	MGL	905

Reaction No. 59901							
5ATP-4 + Sc+3 = Sc+3_5ATP-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.05	Unknown	lgK:6.35 (3SF)	4	MKN	1437

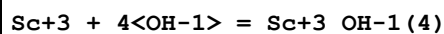
Reaction No. 60375							
Sc+3 + OH-1 = Sc+3_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.0 (2SF)	6	CRV	552
2	25	0	Inf. Dilution	lgK:9.7 (2SF)	4	CRV	2556
3	25	0	Inf. Dilution	lgK:9.3 (2SF)	3	EOD	16981
4	25	0.1	KNO3	lgK:8.98 (3SF)	4	EBP	4880
Note (4): Sc+3 + H2O = Sc+3_OH-1 + H+1 (13.80-4.84)							
5	25	0.1	Unknown	lgK:9.1 (2SF)	5	CRV	2556
6	25	0.5	Unknown	lgK:9.0 (2SF)	5	CRV	2556
7	25	1	Unknown	lgK:8.7 (2SF)	5	CRV	2556
8	10:40	1	Unknown	dH:2. (1SF)	5	CRV	2556

Reaction No. 60376							
Sc+3 + 2<OH-1> = Sc+3_OH-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.3 (3SF)	4	CRV	2556
2	25	0	Inf. Dilution	lgK:17.8 (3SF)	3	EOD	16981
3	25	0.1	Unknown	lgK:17.2 (3SF)	5	CRV	2556
4	25	1	Unknown	lgK:17.4 (3SF)	5	CRV	2556

Reaction No. 60377							
Sc+3 + 3<OH-1> = Sc+3_OH-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.6 (3SF)	5	CRV	2556

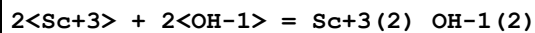
2	25	0	Inf. Dilution	lgK:25.9 (3SF)	4	CRV	2556
3	25	0	Inf. Dilution	lgK:25.4 (3SF)	3	EOD	16981

Reaction No. 60378



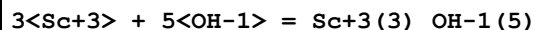
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30. (2SF)	1	CRV	2556

Reaction No. 60379



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.0 (3SF)	4	CRV	2556
2	25	0.1	Unknown	lgK:21.50 (4SF)	5	CRV	2556
3	25	0.5	Unknown	lgK:21.40 (4SF)	5	CRV	2556
4	25	1	Unknown	lgK:21.45 (4SF)	5	CRV	2556
5	10:40	1	Unknown	dH:-12. (2SF)	5	CRV	2556

Reaction No. 60380

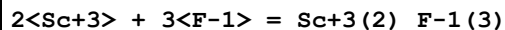


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.8 (3SF)	2	CRV	2556
2	25	0.1	KNO3	lgK:51.43 (4SF)	4	EBP	4880

Note (2): $3\text{<Sc}+3\text{>} + 5\text{<H}_2\text{O>} = \text{Sc}+3(3)\text{_{OH-1(5)}} + 5\text{<H+1>} (13.80*5-17.57)$

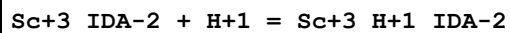
3	25	0.1	Unknown	lgK:51.6 (3SF)	5	CRV	2556
4	25	0.5	Unknown	lgK:51.4 (3SF)	5	CRV	2556
5	25	1	Unknown	lgK:51.5 (3SF)	5	CRV	2556
6	10:40	1	Unknown	dH:-28. (2SF)	5	CRV	2556

Reaction No. 60664



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:19.0 (3SF)	0	CRV	2556

Reaction No. 62494



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Na+1 salt	lgK:0.8 (1SF)	6	CRV	552

Reaction No. 62495							
$\text{Sc}+3_{\text{H}+1_{\text{IDA}-2}} + \text{H}+1 = \text{Sc}+3_{\text{H}+1(2)_{\text{IDA}-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Na+1 salt	lgK:2.9(2SF)	6	CRV	552

Reaction No. 65698							
$\text{SCN}-1 + \text{Sc}+3 = \text{Sc}+3_{\text{SCN}-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:1.87(3SF)	5	MDS	11516
Note (1): Sas T et al., Zhur. Neorg. Khim., 1970, 15, 1255							
2	20	0.5	Unknown	lgK:0.20(2SF)	0	CRV	552
3	20	4	Unknown	lgK:0.8(1SF)	6	CRV	552
4	25	0	Inf. Dilution	lgK:2.4(2SF)	1	EDH	
5	25	0	Inf. Dilution	lgK:2.37(3SF)	2	EAE	

Reaction No. 66429							
$1.5\langle\text{Br}_2(1)\rangle + \text{Sc}(s) = \text{Sc}+3_{\text{Br}-1(3)}(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:125.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:124.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:90.77(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:70.23(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:56.57(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-170.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-743.1(4SF) kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-177.5(4SF)	3	MTD	6422
9	127	0	Inf. Dilution	dH:-188.2(4SF)	2	MTD	6422
10	227	0	Inf. Dilution	dH:-187.8(4SF)	3	MTD	6422
11	327	0	Inf. Dilution	dH:-187.3(4SF)	0	MTD	6422

Reaction No. 66430							
$\text{Sc}(s) + \text{As}(s) + 2\langle\text{O}_2(g)\rangle = \text{Sc}+3_{\text{AsO}_4-3}(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:234.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:232.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:170.2(4SF)	2	ENS	6422

4	227	0	Inf. Dilution	lgK:132.8(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:108.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-319.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-342.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-342.1(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-341.5(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-340.9(4SF)	0	MTD	6422

Reaction No. 66853							
ScOOH(s) + H2O = Sc+3 + 3<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.7(3SF)	4	ENS	773
2	25	1	Unknown	lgK:-30.9(3SF)	4	ENS	773

Reaction No. 67961							
SolochromeVR-3 + Sc+3 = Sc+3_SolochromeVR-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.0(3SF)	4	EDH	7000

Reaction No. 68741							
Sc+3_Malonic-2 + Malonic-2 = Sc+3_Malonic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.12(3SF)	5	MGL	6793
2	25	1	NaClO4	lgK:4.25(3SF)	5	MGL	10137

Reaction No. 68742							
Maleic-2 + Sc+3 = Sc+3_Maleic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.98(3SF)	5	MGL	6793

Reaction No. 68743							
Sc+3_Maleic-2 + Maleic-2 = Sc+3_Maleic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.27(3SF)	5	MGL	6793

Reaction No. 68744							
Itaconic-2 + Sc+3 = Sc+3_Itaconic-2							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.77 (3SF)	5	MGL	6793

Reaction No. 68745

Sc+3_Itaconic-2 + Itaconic-2 = Sc+3_Itaconic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.00 (3SF)	5	MGL	6793

Reaction No. 68746

Sc+3_Glutaric-2 + Glutaric-2 = Sc+3_Glutaric-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.70 (3SF)	5	MGL	6793

Reaction No. 68747

Sc+3_Adipic-2 + Adipic-2 = Sc+3_Adipic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.50 (3SF)	5	MGL	6793

Reaction No. 68748

Sc+3_Succinic-2 + Succinic-2 = Sc+3_Succinic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.88 (3SF)	5	MGL	6793

Reaction No. 68774

Acetic-1 + Sc+3 = Sc+3_Acetic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:3.77 (3SF)	3	CRV	8888

Note (1): Low wgt for internal consistency

2	20	0.1	NaClO4	lgK:3.14 (3SF)	5	MGL	6788
3	25	0	Inf. Dilution	lgK:3.33 (3SF)	3	CRV	8888
4	25	0.1	Unknown	lgK:3.48 (3SF)	2	EOD	31112
5	50	0	Inf. Dilution	lgK:3.10 (3SF)	3	CRV	8888
6	75	0	Inf. Dilution	lgK:2.99 (3SF)	3	CRV	8888
7	100	0	Inf. Dilution	lgK:2.97 (3SF)	3	CRV	8888
8	125	0	Inf. Dilution	lgK:3.02 (3SF)	3	CRV	8888
9	150	0	Inf. Dilution	lgK:3.12 (3SF)	3	CRV	8888
10	175	0	Inf. Dilution	lgK:3.27 (3SF)	3	CRV	8888

11	200	0	Inf. Dilution	lgK:3.45 (3SF)	3	CRV	8888
12	225	0	Inf. Dilution	lgK:3.67 (3SF)	3	CRV	8888
13	250	0	Inf. Dilution	lgK:3.93 (3SF)	3	CRV	8888
14	275	0	Inf. Dilution	lgK:4.24 (3SF)	3	CRV	8888
15	300	0	Inf. Dilution	lgK:4.61 (3SF)	3	CRV	8888
16	325	0	Inf. Dilution	lgK:5.06 (3SF)	3	CRV	8888
17	350	0	Inf. Dilution	lgK:5.54 (3SF)	3	CRV	8888

Reaction No. 68777							
Sc+3 + H+1(-1)_Tartaric-2 + H+1 = Sc+3_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:17.60 (4SF)	0	MGL	22560
Note (1): Wrong NDP							

Reaction No. 68778							
Sc+3 + H+1_Tartaric-2 = Sc+3_Tartaric-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:3.20 (3SF)	4	MGL	6788

Reaction No. 68782							
Sc+3 + H+1(-1)_Tartaric-2 = Sc+3_H+1(-1)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:17.60 (4SF)	0	MGL	22560
Note (1): Wrong NDP							

Reaction No. 68783							
Sc+3 + H+1_Citric-3 = Sc+3_Citric-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:5.62 (3SF)	0	MGL	6788
2	25	0	Inf. Dilution	lgK:2.5 (2SF)	1	ELC	

Reaction No. 68784							
Sc+3_Citric-3 + H2O = Sc+3_OH-1_Citric-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:21.62 (4SF)	0	MGL	6788
2	25	0	Inf. Dilution	lgK:-0.6 (1SF)	1	ELC	

Reaction No. 69164							
Sc+3 + H+1_4NitrPhenol-1 = Sc+3_H+1_4NitrPhenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:5.2(0.1SD)	5	MGL	4872

Reaction No. 69166							
Sc+3 + H+1_2NitrPhenol-1 = Sc+3_H+1_2NitrPhenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:5.0(0.1SD)	5	MGL	4872

Reaction No. 69168							
Sc+3 + H+1_24DiNitrphenol-1 = Sc+3_H+1_24DiNitrphenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:2.(0.3SD)	5	MGL	4872

Reaction No. 69170							
Sc+3 + H+1_26DiNitrphenol-1 = Sc+3_H+1_26DiNitrphenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:2.6(0.1SD)	5	MGL	4872

Reaction No. 69345							
Sc+3 + 2<XylenolOrange-6> = Sc+3_XylenolOrange-6(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.6	KNO3	lgK:10.8(3SF)	5	MPS	7149
2	20	0.7	KNO3	lgK:10.84(4SF)	5	MPS	7149
3	20	0.8	KNO3	lgK:10.94(4SF)	5	MPS	7149
4	20	0.9	KNO3	lgK:11.02(4SF)	5	MPS	7149
5	20	1	KNO3	lgK:11.06(4SF)	5	MPS	7149

Reaction No. 69346							
Sc+3_XylenolOrange-6(2) + Sc+3 = Sc+3(2)_XylenolOrange-6(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.6	KNO3	lgK:3.87(3SF)	5	MPS	7149
2	20	0.7	KNO3	lgK:3.94(3SF)	5	MPS	7149
3	20	0.8	KNO3	lgK:3.98(3SF)	5	MPS	7149
4	20	0.9	KNO3	lgK:4.04(3SF)	5	MPS	7149
5	20	1	KNO3	lgK:4.09(3SF)	5	MPS	7149

Reaction No. 69642							
Sc+3 + NH3 = Sc+3_NH3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.7 (4SF)	1	EOR	
2	25	5	Unknown	lgK:0.7 (1SF)	4	EES	1548

Reaction No. 70724							
Glycolic-1 + Sc+3 = Sc+3_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.40 (3SF)	5	MGL	7703

Reaction No. 70766							
2OH2MePropan-1 + Sc+3 = Sc+3_2OH2MePropan-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.84 (3SF)	5	MQH	7703

Reaction No. 71633							
Sc+3 + 1.5<H2(g)> + 3<e-1> = ScH3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.75 (3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:485.0 (4SF) kJ	3	CRV	7761

Reaction No. 71634							
Sc+2 + 2<e-1> = Sc(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.0 (2SF)	0	CRV	7761
Note (2): Low wgt for inter-reaction consistency							
3	25	0	Inf. Dilution	dH:374.4 (4SF) kJ	0	CRV	7761

Reaction No. 71635							
Sc+3_OH-1 + H+1 + 3<e-1> = Sc(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-98.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.01 (3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:602.5 (4SF) kJ	2	CRV	7761

Reaction No. 71636							
$\text{Sc}+3(2)\text{OH-1}(2) + 2\text{H}^+ + 6\text{e}^- = 2\text{Sc(s)} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -199.3(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -2.03(3\text{SF})$	0	CRV	7761

Reaction No. 71637							
$\text{Sc}+3 + \text{e}^- = \text{Sc}+2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.3(2\text{SF})$	3	CRV	7761
2	25	0	Inf. Dilution	$\text{dH}: 267.9(4\text{SF}) \text{ kJ}$	3	CRV	7761

Reaction No. 71638							
$\text{Sc}+3\text{OH-1}(4) + 3\text{e}^- = \text{Sc(s)} + 4\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -132.7(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -2.68(3\text{SF})$	0	CRV	7761
3	25	0	Inf. Dilution	$\text{dH}: 675.6(4\text{SF}) \text{ kJ}$	2	CRV	7761

Reaction No. 71639							
$\text{Sc}+3\text{OH-1}(3)\text{_(am.,s)} + 3\text{e}^- = \text{Sc(s)} + 3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.69(3\text{SF})$	4	CRV	7761
2	25	0	Inf. Dilution	$\text{dH}: 691.5(4\text{SF}) \text{ kJ}$	3	CRV	7761

Reaction No. 71640							
$\text{Sc}+3\text{OH-1}(3)\text{_(s)} + 3\text{e}^- = \text{Sc(s)} + 3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -133.1(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -2.72(3\text{SF})$	0	CRV	7761
3	25	0	Inf. Dilution	$E: -2.60(3\text{SF})$	0	CRV	22551
4	25	0	Inf. Dilution	$\text{dH}: 704.5(4\text{SF}) \text{ kJ}$	1	CRV	7761

Reaction No. 71641							
$\text{Sc}_2\text{O}_3(\text{s}) + 3\text{H}_2\text{O} + 6\text{e}^- = 2\text{Sc(s)} + 6\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.742(4\text{SF})$	5	CRV	7761

2	25	0	Inf. Dilution	dH:1387.(4SF)kJ	5	CRV	7761
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Reaction No. 71642							
$\text{ScOOH(s)} + \text{H}_2\text{O} + 3\text{e-1} = \text{Sc(s)} + 3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.76(3SF)	0	CRV	7761
Note (1): Low wgt for inter-reaction consistency							
2	25	0	Inf. Dilution	dH:704.0(4SF)kJ	0	CRV	7761

Reaction No. 72273							
$\text{ScOOH(s)} + 3\text{H+1} = \text{Sc+3} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:9.4(0.1SD)	0	CRV	6478
3	25	0	Inf. Dilution	dH:-25.4(0.9SD)	2	CRV	6478

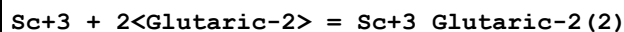
Reaction No. 74026							
$\text{Sc+3}_{\text{OH-1(3)}}(\text{s}) = \text{Sc+3} + 3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-29.4(3SF)	0	EOD	16981
2	25	0	Inf. Dilution	lgK:-30.4(3SF)	5	CRV	22551

Reaction No. 75031							
$\text{CrO4-2} + \text{Sc+3} = \text{Sc+3}_{\text{CrO4-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.01	0	Inf. Dilution	lgK:7.2654(5SF)	3	EOD	22958
2	25	0	Inf. Dilution	lgK:7.7464(5SF)	3	EOD	22958
3	60	0	Inf. Dilution	lgK:8.5515(5SF)	3	EOD	22958
4	100	0	Inf. Dilution	lgK:9.5252(5SF)	3	EOD	22958
5	150	0	Inf. Dilution	lgK:10.7850(6SF)	3	EOD	22958
6	200	0	Inf. Dilution	lgK:12.1190(6SF)	3	EOD	22958

Reaction No. 75947							
$\text{Sc+3}_{\text{Oxalic-2}} + \text{Oxalic-2} = \text{Sc+3}_{\text{Oxalic-2(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:5.55(3SF)	5	MDS	11516
Note (1): Gardhammar G et al., Acta Chem. Scand., 1971, 25, 158							

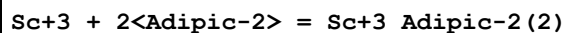
2	25	0	Inf. Dilution	lgK:6.11 (3SF)	2	EAE	
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Reaction No. 76324



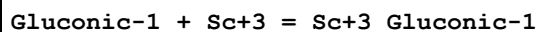
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.91 (3SF)	5	MGL	6793

Reaction No. 76335



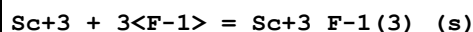
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.75 (3SF)	5	MGL	6793

Reaction No. 77375



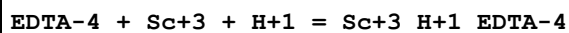
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:4.21 (3SF)	4	CRV	7271

Reaction No. 77657



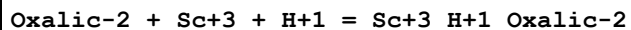
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.33 (4SF)	1	ELC	

Reaction No. 77978



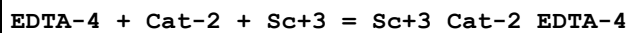
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.8 (4SF)	1	ELC	

Reaction No. 78149



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.12 (4SF)	1	ELC	

Reaction No. 78241



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.28 (4SF)	1	ELC	

Reaction No. 78462

2<Sc+3> + PO4-3 + H+1 = Sc+3(2)_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.2(4SF)	1	ELC	

Reaction No. 78472							
Sc+3 + PO4-3 + 2<H+1> = Sc+3_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.24(4SF)	1	ELC	

Reaction No. 78540							
Sc+3 + 2<BrO3-1> = Sc+3_BrO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.222(4SF)	1	ELC	

Reaction No. 78622							
EDTA-4 + Asn-1 + Sc+3 = Sc+3_Asn-1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.72(4SF)	1	ELC	

Reaction No. 78925							
Sc+3 + 3<Br-1> = Sc+3_Br-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.42(4SF)	1	ELC	

Reaction No. 78926							
Sc+3 + AsO4-3 = Sc+3_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.91(4SF)	1	ELC	

Reaction No. 79158							
Sc+3 + 3<OH-1> = Sc+3_OH-1(3)_(am.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.75(4SF)	1	ELC	

Reaction No. 79471							
Sc+3 + ClO4-1 = Sc+3_ClO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.(1SF)	1	EES	

Reaction No. 80692							
Sc+3_OH-1 + H2O = Sc+3_OH-1(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.4(2SF)	1	ELC	
2	25	0	Unknown	lgK:-9.7(2SF)	0	EOD	31393

Reaction No. 80693							
Sc+3_OH-1(2) + H2O = Sc+3_OH-1(3) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.4(2SF)	1	ELC	
2	25	0	Unknown	lgK:-18.1(3SF)	0	EOD	31393

Reaction No. 80694							
Sc+3_OH-1(3) + H2O = Sc+3_OH-1(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.9(2SF)	1	ELC	
2	25	0	Unknown	lgK:-26.(2SF)	0	EOD	31393

Reaction No. 80695							
[12]N4:Acet*4-4 + Sc+3 = Sc+3_[12]N4:Acet*4-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:29.3(0.2SD)	1	MPH	31393
2	25	0.1	NaCl	lgK:30.79(0.05SD)	1	EOD	31393

Reaction No. 80696							
Sc+3_[12]N4:Acet*4-4 + H+1 = Sc+3_H+1_[12]N4:Acet*4-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:1.36(0.05SD)	2	EOD	31393

Reaction No. 80697							
Sc+3_DTPA-5 + H+1 = Sc+3_H+1_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:1.00(0.05SD)	2	EOD	31393

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EAC	Activity Coefficient
EAE	Automated extrapolation
EBP	By Proxy
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
EMU	Method unknown
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
EVK	Variation of lgK with T
MAG	Silver electrode e.m.f.
MAX	Anion exchange

MCE	Cation exchange
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MGL	Glass electrode
MHG	Emf with amalgam electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MMR	Nuclear magnetic resonance
MPH	pH method, not specified
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MQH	Quinhydrone electrode
MRX	Emf with redox electrode
MSC	Miscellaneous
MSP	Spectroscopy
MTD	Temperature difference

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